If (t) t is a family of probability measures on / , increasing for , or , there is a Mark...

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Editorial note. All delivered mathematical text is the author's own, copied from the frozen source sources/markov-quantile-jep-2022-177/source0.tex; no statement or proof step was rewritten, reordered or removed. References inside the span are handed to the assembler as navigation stubs; load-bearing status is recorded per stub.

Editorial scope. Single themBJ card cut from doi:10.5802/jep.177 (The Markov-quantile process attached to a family of marginals); statement and proof are the author own text at source lines 440-441 and 868-882.

Source reference. Original theorem 2.21: sources/markov-quantile-jep-2022-177/source0.tex, source lines 833–833. navigation stub only: the target lives outside the counted span in the same article; whether it is load-bearing is judged per occurrence

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Source reference. Original theorem 3.24: sources/markov-quantile-jep-2022-177/source0.tex, source lines 1173–1173. navigation stub only: the target lives outside the counted span in the same article; whether it is load-bearing is judged per occurrence

Theorem C. *If $(\mu_t)_{t\in\mathbb{R}}$ is a family of probability measures on $\mathbb{R}$, increasing for $\preceq_C$, $\preceq_{C,\mathrm{sto}}$ or $\preceq_{\mathrm{sto}}$, there is a Markov measure $P \in \mathrm{Marg}((\mu_t)_t)$ which respectively is a martingale, is a submartingale or has increasing couplings.*

Proof of Theorem C. Take $(\mu_t)_t \in \mathcal{P}(\mathbb{R})^{\mathbb{R}}$, increasing for $\preceq_C$ (case (a)), $\preceq_{C,\mathrm{sto}}$ (case (b)) or for $\preceq_{\mathrm{sto}}$ (case (c)) to prove the corresponding cases of Theorem C. In the sketch of proof of Theorem 2.21 given in Remark 2.22, replace $\mathcal{N}^{\mathrm{LK}}_{s,t}$ by $\mathcal{N}^{\mathrm{IK}}_{s,t}$ and introduce, similarly as defined in Remark 2.22 for cases (a) and (b), the spaces $\mathcal{N}'^{\mathrm{IK}}_{s,t}$—equal to $\{P \in \mathcal{N}^{\mathrm{IK}}_{s,t} : P(\{(x,y) \in \mathbb{R}^2 : x \leqslant y\}) = 1\}$ in case (c).

Properties (2.15)–(2.15) of Theorem 2.15 are satisfied by $(\mathcal{N}^{\mathrm{IK}}_{s,t})_{s<t}$: (2.15) by definition, (2.15) by Lemma 2.24, (2.15) is straightforward and (2.15) by Lemma 2.23. By Remark 2.18(2.18) we are left with showing (2.15), (2.15) and (2.15) for $(\mathcal{N}'^{\mathrm{IK}}_{s,t})_{s<t}$. Plainly, the conditions defining the $\mathcal{N}'^{\mathrm{IK}}_{s,t}$ as subspaces are closed and stable by composition, (2.15) and (2.15) follow. For (2.15), in our three cases:

a. By [1, §3.1], if $P \in \mathrm{Marg}(\mu_s, \mu_t)$ is a martingale transport plan, then $P \in \mathcal{N}^{\mathrm{LK}}_{s,t} \Leftrightarrow P \in \mathcal{N}^{\mathrm{IK}}_{s,t}$. Therefore $\mathcal{N}'^{\mathrm{IK}}_{s,t} \neq \varnothing$ if and only if $\mathcal{N}'^{\mathrm{LK}}_{s,t} \neq \varnothing$, which Kellerer proved.

b. The element of $\mathcal{N}'^{\mathrm{LK}}_{s,t}$ Kellerer built in [2, Def. 7 & Th. 2] is in $\mathcal{N}'^{\mathrm{IK}}_{s,t}$ when $\mu_s \preceq_C \mu_t$.

c. As explained in Remark 3.24, $\mathfrak{Q}(\mu_s, \mu_t) \in \mathcal{N}'^{\mathrm{IK}}_{s,t}$. □

References

[1] M. Beiglböck, M. Huesmann & F. Stebegg - "Root to Kellerer", in Séminaire de Probabilités XLVIII, Lect. Notes in Math., vol. 2168, Springer, Cham, 2016, p. 1-12

[2] H. G. Kellerer - "Markov-Komposition und eine Anwendung auf Martingale", Math. Ann. 198 (1972), p. 99-122